

Predicting Breast Cancer Diagnosis with Supervised Classification and Unsupervised Clustering

Dixin Zhang

CIS-435 Practical Data Science Using Machine Learning

Professor Sunil Kakade

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1. Business Problem and Machine Learning Process

The primary goal of this project is to aid in earlier and more accurate detection of malignant tumors based on diagnostic measurements. In healthcare settings, this is important because a false negative in classifying a malignant tumor as benign can delay treatment and negatively impact patient outcomes. Thus, model evaluation should prioritize metrics that capture rates of clinical risk rather than just general accuracy.

The project to solve this problem will instead follow the typical workflow of machine learning. First, the prediction goal is defined as a binary classification problem. Then, the data set is explored to determine the types of features present, if the classes are well-balanced. The data are then prepared by removing non-predictive identifiers and applying scaling where appropriate. Multiple supervised classification models are trained and compared using consistent validation procedures. After evaluation, selected models are improved through hyperparameter tuning. Finally, unsupervised clustering is used to identify natural groupings in the data and examine how these segments relate to malignancy risk.

2. Machine Learning Applications

This case study relates to several established applications of machine learning in clinical decision support.

2.1 Computer-aided cancer detection using medical imaging: Deep learning can classify medical images to improve early cancer detection and support clinician consistency (Jeyaraj & Nadar, 2019).

2.2 Early Alzheimer's disease diagnosis from brain networks: Models can combine imaging-derived network features and clinical variables to identify early-stage impairment (Ju et al., 2019).

2.3 Early Parkinson's disease diagnosis using machine learning: Predictive models can assist early detection of neurodegenerative disease using clinical/biomedical features (Senturk, 2020).

3. Machine Learning Algorithms

3.1 Logistic Regression. It is efficient to train, provides probabilistic outputs, and offers relatively good interpretability. However, it assumes a linear decision boundary, which may limit performance if relationships are highly non-linear.

3.2 Decision Trees. Generating human-readable decision rules and can model non-linear patterns without requiring feature scaling. Their main limitation is a tendency to overfit unless depth and complexity are carefully controlled.

3.3 Naive Bayes. While it can perform well as a baseline, its assumption of conditional independence among predictors may not hold for correlated medical features.

3.4 Multilayer Perceptron (MLP) neural networks. Flexible and capable of modeling complex non-linear relationships. However, they are less interpretable than simpler models and require careful tuning to avoid overfitting.

3.5 KMeans clustering. Used for unsupervised segmentation. It is simple and effective for discovering structure without labels, but it requires choosing the number of clusters in advance and is sensitive to feature scaling and outliers.

4. Data Understanding and Preprocessing

The dataset contains 569 observations and 21 numeric diagnostic features. In addition, there is an identifier column and a binary diagnosis label. The target variable is moderately imbalanced, with 62.7% benign cases and 37.3% malignant cases. This imbalance explains why recall and ROC-AUC are especially important when reporting performance measures.

```
Target value counts:

diagnosis
0      357
1      212
Name: count, dtype: int64

Target proportions:

diagnosis
0      0.627417
1      0.372583
Name: proportion, dtype: float64
```

4.1 Target proportions

After the initial data checks, there were no missing values. Since the identifier column does not provide any predictive information, it was removed. The data were split into training and testing sets with an 80/20 stratified split to maintain the proportions of the classes. Feature scaling was used for all models except the Decision Tree, as it is not sensitive to the magnitude of the features. To further improve reliability of the model evaluation, a stratified 5-fold cross-validation was used, which is more robust than a single split.

5. Metrics Used to Analyze Model Performance

There are several metrics that capture different aspects of model performance.

Recall (sensitivity) is the proportion of malignant cases that are correctly identified and is especially relevant in screening tasks. Precision is the proportion of predicted malignant cases that are actually malignant, which can affect unnecessary follow-up examinations. The F1-score is the harmonic mean of precision and recall. ROC-AUC evaluates how well a model discriminates between the two classes across all classification thresholds.

These metrics are commonly used in medical machine learning research to evaluate classification quality in imbalanced settings (Huang et al., 2012; Salvatore et al., 2015).

6. Interpreting and Analyzing Results

6.1 Supervised Learning Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.947	0.950	0.905	0.927	0.988
Neural Network (MLP)	0.921	0.923	0.857	0.889	0.982
Naive Bayes	0.895	0.895	0.810	0.850	0.980
Decision Tree	0.921	0.902	0.881	0.892	0.913

6.1 Baseline model performance (test set)

At baseline, Logistic Regression produced the strongest overall discrimination (highest ROC-AUC). Although MLP and Naive Bayes also achieved high ROC-AUC, Logistic Regression provided the best combination of recall and stability, which is valuable for screening settings where missing malignant cases are costly.

Cross-validation (5-fold, stratified) further supported this conclusion. Logistic Regression achieved mean ROC-AUC 0.992 ± 0.005 and accuracy 0.965 ± 0.004 , outperforming the other models on average and showing low variability across folds.

After tuning, Logistic Regression improved accuracy and achieved perfect precision (1.000) on the test set while keeping recall at 0.905. In practical terms, this tuned model produced no false positives but still had four false negatives, indicating that while predictions were highly trustworthy when flagged “malignant,” a small number of malignant cases were still missed. For healthcare use, this result is strong but still motivates threshold tuning if the operational priority is maximizing sensitivity.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Tuned	0.965	1.000	0.905	0.950	0.987
Logistic Regression					
Tuned MLP	0.947	0.974	0.881	0.925	0.984
Tuned Decision Tree	0.921	0.884	0.905	0.894	0.910

6.2 Tuned model performance (test set)

Feature analysis (coefficients and permutation importance) indicated that a small set of diagnostic measurements contributed disproportionately to prediction quality, supporting the idea that interpretable models can still deliver strong performance when the features are informative.

6.2 Unsupervised Learning Results (KMeans)

KMeans clustering was used to identify natural patient segments without labels. The optimal number of clusters was selected by comparing inertia (elbow) and silhouette score across $k = 2$ to

10. The best silhouette score occurred at $k = 2$ (silhouette ≈ 0.351), supporting a two-segment structure.

	k	inertia	silhouette
0	2	8304.409577	0.351388
1	3	7105.122756	0.311543
2	4	6398.365770	0.282938
3	5	5877.480102	0.204968
4	6	5447.271539	0.189563
5	7	5087.886078	0.194405
6	8	4864.601213	0.167711
7	9	4693.707894	0.137877
8	10	4481.556868	0.145021

6.3 Best k (by silhouette)

This finding suggests that a high-risk feature profile cluster and low-risk cluster may be present in the data set. A visual representation of a PCA (first two components explained $\sim 65.9\%$ of the variance) demonstrated that the segments were substantially separated in projected feature space. From a business standpoint, clustering can help risk stratification, operational planning, and targeted review processes (e.g., triage of cluster 0 for additional clinical review).

7. Recommended Steps for Performance Optimization

The clinical utility of the models could potentially be improved with several steps. The current classification threshold may be adjusted to improve sensitivity at the cost of additional false positives, which may be acceptable in a screening setting. Probability calibration would improve the fidelity of risk estimates if the predictions were used for triage. Additional evaluation with more rigorous cross-validation (e.g., nested cross-validation) would decrease the

risk of evaluation bias. External validation on other populations would be necessary before any deployment.

8. Conclusion

This work illustrates that supervised classification models can predict breast cancer diagnosis using structured diagnostic features. Logistic Regression performed strongly and was stable across validation sets while maintaining interpretability; it therefore may be a practical tool for use as clinical decision support. Unsupervised clustering techniques provided additional insight by identifying meaningful patient segments with respect to malignancy risk.

References

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